

Experiment 1 – Microsoft Threat Modeling

Aim

To create a threat model for a given system using Microsoft Threat Modeling Tool, identify threats using STRIDE, and define suitable mitigations.

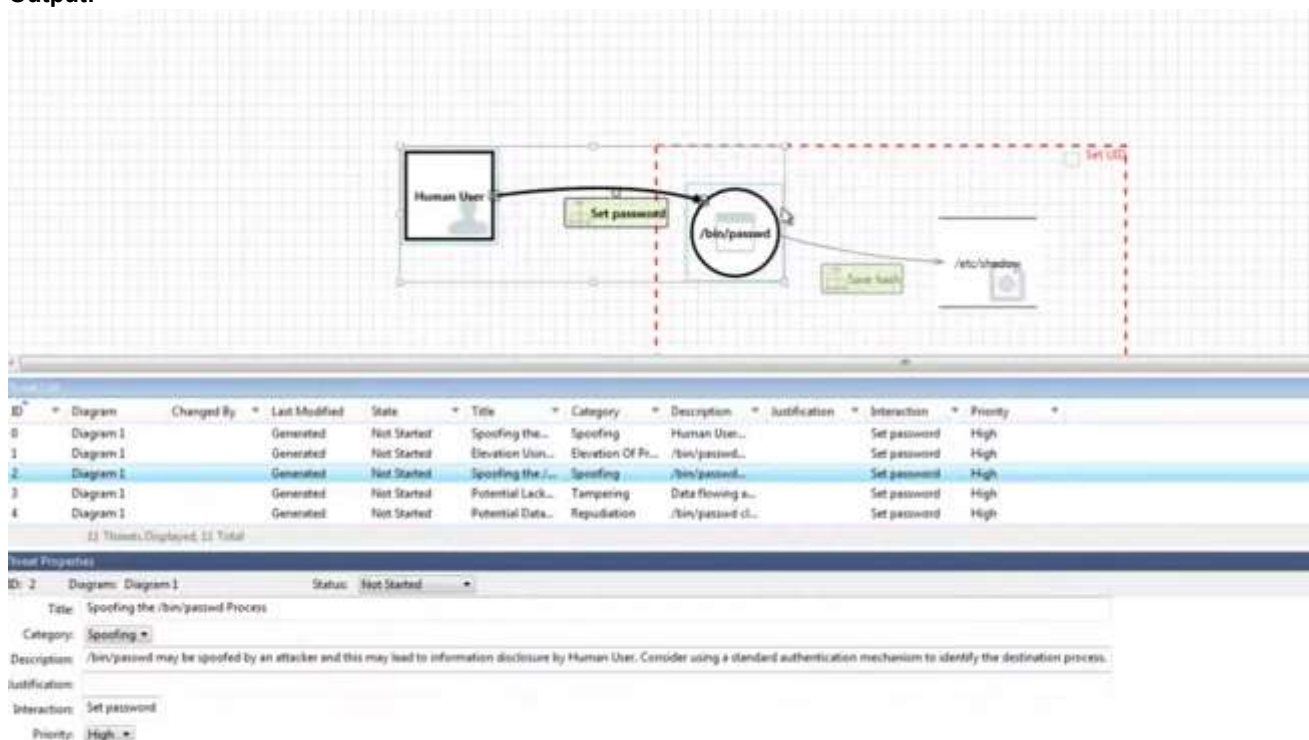
Tools:

Microsoft Threat modelling tool

Procedure

1. Open Microsoft Threat Modeling Tool.
2. Create a new threat model and select the available SDL/Threat Modeling template.
3. Read the given scenario and identify external entities, processes, data stores, data flows and trust boundaries.
4. Add and rename the required elements on the diagram.
5. Connect components using Data Flow arrows and add Trust Boundaries where required.
6. Save the model and select Analyze.
7. Review STRIDE threats: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service and Elevation of Privilege.
8. Set threat priority/status and add appropriate mitigations.
9. Generate the report using Reports → Create Full Report.

Output:



Result:

Threat model successfully created, STRIDE security threats identified and prioritized, mitigations assigned, and the full threat modeling report generated